FATURA ANALIZ

Düzce University
Faculty of Engineering

PRESENTATION 01 — TERM KICK-OFF

An AI-Supported Expense & Invoice *Analysis System*

A mobile platform that turns a photo of a receipt into a categorized expense — powered by the Google Gemini 2.5 Flash multimodal model.

Student
Uğur Avşar · 211015040

Supervisor
Dr. Büşra Takgil

Course
CE498 · Project Design

Term
2025–2026 Spring

TODAY'S AGENDA

01 **The problem**

Why consistent expense tracking breaks down — and where today's apps fall short.

02 **Aim & research questions**

What this project sets out to prove, framed as four testable questions.

03 **Proposed solution & scope**

A four-component system, its boundaries, and known limitations.

04 **Methodology & plan**

An iterative, layer-by-layer build plan and the term's deliverables.

01 — PROBLEM DEFINITION

Receipts pile up unrecorded, invoice photos get lost, and payments scatter across apps — so a *coherent* picture of one's spending is almost impossible to build.

Tracking expenses consistently is fragmented and unsustainable in practice — the friction of manual entry is the real barrier, not a lack of motivation.

Rule-based logic

Tools like Mint and Money Manager rely on fixed rules that fail to adapt to diverse users and varied invoice formats.

Text-only processing

Most systems read transaction text only — photographed receipts and invoices cannot be processed automatically.

Tightly-coupled AI

Where AI exists, it is embedded in the backend — hard to update or replace, and rarely multimodal.

> The literature has barely explored **multimodal models that read image and text together** for invoice analysis on mobile.

From data-entry operator to *passive reviewer.*

Users submit an invoice image and receive an auto-generated expense record back — drastically cutting the effort of consistent tracking.

→ **Practical**

Removes the manual-entry dependency that limits existing finance apps.

→ **Technical**

A concrete full-stack implementation of Gemini 2.5 Flash in a mobile app.

→ **Academic**

Empirical evidence for an underexplored area: vision-language models for invoice analysis.

RQ1 **Can Gemini 2.5 Flash** accurately extract structured financial information — total, date, merchant — from diverse invoice images?

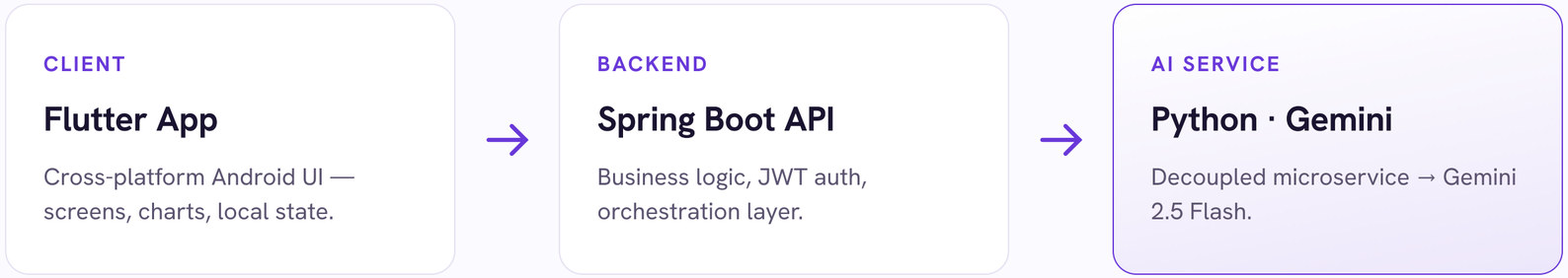
RQ2 **Does AI-based automatic categorization** meaningfully reduce the manual input burden on users?

RQ3 **Does a decoupled Python microservice** allow AI integration without harming system stability and response times?

RQ4 **How does the system perform across** end-to-end scenarios combining AI analysis, manual management, and reporting?

03 — PROPOSED SOLUTION

One platform, four cooperating components.



> A PostgreSQL database stores user accounts, expenses, and budget limits — isolated per user.

Secure access

JWT-based registration & login with per-user data isolation and email verification.

Expense management

Manual entry, listing, editing and deletion across ten predefined categories.

AI-assisted import

Invoice-photo analysis and CSV/XLSX bulk import with automatic categorization.

Statistics

Totals, daily average, category distribution and seven-day spending trend.

Budget & comparison

Per-category limits, monthly comparison and AI commentary.

Reporting

PDF monthly report, dark mode, and Turkish / English localization.

01

Dependent on an external AI service

Invoice analysis relies entirely on the Gemini 2.5 Flash API — no offline fallback or locally-hosted model.

02

Limited invoice-format coverage

Accuracy is evaluated on a manually prepared test set, not the full diversity of industries, countries and languages.

03

Single currency & untracked corrections

Turkish Lira only; when a user overrides an AI category, the original prediction is not retained for analysis.

04 — DEVELOPMENT METHODOLOGY

Iterative & incremental — one layer at a time.

ITERATION 1

Foundation

Backend infrastructure, database schema, and JWT authentication.

ITERATION 2

Core CRUD

Expense management endpoints and the basic mobile client.

ITERATION 3

AI integration

Python microservice, Gemini invoice analysis and CSV import.

ITERATION 4

Analytics & polish

Statistics tabs, budgets, profile, dark mode & PDF report.

04 — PLANNED TECHNOLOGY STACK

MOBILE

Flutter · Dart

Single codebase, Skia/Impeller rendering for pixel-consistent charts.

BACKEND

Spring Boot 3

Java · Spring Security, Spring Data JPA, embedded Tomcat.

DATABASE

PostgreSQL

ACID transactions, mature indexing, JPA / Hibernate ORM.

AI SERVICE

Python · Flask

Decoupled microservice wrapping Google Gemini 2.5 Flash.

JWT · BCrypt · REST API · fl_chart · image_picker · pandas · pdf / printing · intl localization

4

Integrated components delivered & tested end-to-end

3×

AI modes: photo analysis, CSV import, statistical commentary

10

Expense categories shared across every layer of the app

> A functional, secure and extensible system — validated against the four research questions in the final report.

PRESENTATION 01 — TERM KICK-OFF

Thank you.
Questions & discussion.

STUDENT
Uğur Avşar · 211015040

SUPERVISOR
Dr. Büşra Takgil

COURSE
CE498 · 2025–2026 Spring